

PAULA MAGDY

Software Engineer

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PROFILE

Full-Stack Software Engineer who ships production systems end-to-end across multiple stacks — TypeScript (Next.js/NestJS), PHP (Laravel), and Python (Django/Flask) — on AWS, Azure, and Huawei Cloud. AWS Certified Solutions Architect who owns delivery from data modeling and API architecture through CI/CD, monitoring, and security, shipping government-grade and enterprise platforms that serve thousands of concurrent users.

PROFESSIONAL EXPERIENCE

Challenge Group — Full-Stack Software Engineer

Oct 2025 — Present

Cairo, Egypt · AWS/Huawei Cloud partner delivering production platforms for government and enterprise clients across Egypt, SA and the UAE.

▪ **Make it in the Emirates Awards — MoIAT** — a government-grade awards platform for the UAE Ministry of Industry & Advanced Technology, supporting 5,000+ concurrent users.

- Owned delivery from schema to deploy: domain-driven Next.js/NestJS modules over a PostgreSQL/Prisma schema with targeted indexing and query-tuning that cut API latency by ~35% under production load.
- Built the AWS infrastructure (VPC, EC2, RDS, S3, CloudFront, IAM, Secrets Manager) and a container-first Docker → ECR → ECS pipeline (CodeBuild/CodePipeline), reducing deploys from ~20 minutes to under 5.
- Hardened the platform with JWT auth, RBAC, structured validation, and API hardening; added CloudWatch monitoring and structured logging, a full test pyramid, and SEO via SSR, structured metadata, and schema markup.

Stack: Next.js, NestJS, TypeScript, PostgreSQL, Prisma, AWS, Docker.

▪ **LawMate — Legal Services Platform** — End-to-end API-first legal services platform built with Laravel and Filament.

- Built a Laravel backend (PHP, Sanctum authentication, attribute-based routing, self-validating DTOs) powering a request-and-bid workflow connecting clients with lawyers, including escalation handling and an in-app wallet with Paymob payments and withdrawals.
- Architected the platform for real-time chat, voice/SMS (Laravel Reverb WebSockets, Twilio), FCM mobile push notifications, S3-backed media storage, bilingual (EN/AR) content, a role-based Filament (Livewire) admin panel, and automated testing with Pest and PHPUnit.

Stack: Laravel, PHP, MySQL, Redis, Filament (Livewire), Sanctum, Reverb (WebSockets), Twilio, Paymob, AWS S3, Firebase Cloud Messaging (FCM), Pest, PHPUnit.

▪ **Egyptian Schools League — Competition Engine** — a configurable rules engine that replaced manual spreadsheet workflows.

- Engineered a modular, configurable rules engine handling team onboarding, scoring, and ranking logic.
- Built the full-stack application on Next.js and NestJS over a PostgreSQL/Prisma data model, with real-time leaderboards that update live as results are recorded.

Stack: Next.js, React, TypeScript, NestJS, PostgreSQL, Prisma, Docker.

▪ **Sabeel — Mobile Backend** — high-traffic mobile product.

- Hardened the API surface and cut response latency for a high-traffic mobile product.
- Provisioned and maintained the AWS production infrastructure — EC2, S3, IAM, and RDS — behind a CI/CD release pipeline.

Stack: Laravel, PHP, MySQL, PHPUnit, AWS, CI/CD.

▪ **WhatsApp AI Chatbot Automation** — serverless customer-support automation.

- Architected an AI chatbot on Amazon Lex, integrated with Twilio, to automate customer-support conversations and eliminate manual triage.
- Productionized it as a serverless application on AWS Lambda, using Secrets Manager for credentials and CloudWatch for logging and monitoring.

Stack: Python, AWS Lambda, Amazon Lex, Twilio API, Secrets Manager, CloudWatch.

▪ **Eltanfeethi — Azure Delivery** — client-facing application.

- Led the Azure infrastructure for a client-facing application, including Azure API Management.
- Built and owned the CI/CD release pipelines, carrying the app through to production deployment.

Stack: Microsoft Azure, Azure API Management, CI/CD.

Information Technology Institute (ITI) — Full-Stack Engineer

Nov 2024 – Jun 2025

Cairo, Egypt · Government-sponsored intensive software-engineering program (~1,200 hours)

- Built full-stack applications across two backend ecosystems — Node.js/Express and Python (Django, Flask) — with React front-ends, secure REST APIs, structured authentication flows, and role-based access control.
- Modeled both relational (SQL) and NoSQL databases, applying normalization and indexing to keep queries efficient as datasets grew.
- Containerized services with Docker and deployed to AWS, applying system-design fundamentals and working in Agile sprints from design through delivery.

Freelance — Front-End Developer

Oct 2022 – Oct 2024

Cairo, Egypt

- Built and shipped a production-ready e-commerce storefront in React — a responsive, cross-device UI wired to REST APIs for the product catalog and cart.
- Architected a reusable component library with centralized state via React's Context API, removing prop-drilling and keeping catalog and cart state consistent as the product range and traffic grew.
- Improved rendering performance and cut page-load times through render optimization and efficient asset loading, keeping large product listings smooth as the catalog scaled.

SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL, C++

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, CSS, HTML, Livewire, Filament, Blade

Backend: Node.js, NestJS, Express.js, Laravel, Django, Flask, REST APIs, WebSockets, Reverb, Serverless, JWT, RBAC

Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, Eloquent ORM, Query Optimization

Cloud & DevOps: AWS, Huawei Cloud, Microsoft Azure, Docker, Kubernetes, CI/CD

Testing & Practices: Vitest, Jest, PHPUnit, Pest, React Testing Library, Playwright, Domain-Driven Design, Agile/Scrum

CERTIFICATIONS

Huawei HCCDP – Solution Architectures

Jun 2026

Professional-tier certification in end-to-end solution design on Huawei Cloud — architecting for high availability, security, and cost efficiency across the platform's compute (ECS), storage (OBS), networking (VPC/ELB), and managed-database services.

AWS Certified Solutions Architect – Associate (SAA-C03)

Jun 2026

Validates designing fault-tolerant, scalable, and secure AWS architectures — VPC networking, EC2/Lambda compute, S3 and RDS/DynamoDB data layers, IAM, and multi-AZ high availability and disaster recovery.

Huawei HCCDA – Tech Essentials

May 2026

Associate-level foundation in Huawei Cloud — core compute, storage, and networking services, application deployment, and the platform's developer toolchain.

AWS Certified Cloud Practitioner

Apr 2026

Foundational AWS certification spanning cloud concepts and economics, the shared-responsibility security model, pricing and billing, and the platform's core service categories.

EDUCATION

B.Sc., Computer Science and Information Systems

2018 – 2022

October 6 University — Giza, Egypt

- Relevant coursework: Algorithms, Data Structures, Cloud Computing, Database Systems, Software Engineering.
- Graduation project (grade A): Learning Management System (Node.js, Express, MongoDB).